IIIT BHOPAL METAVERSE

AN INTERACTIVE VIRTUAL CAMPUS

&

REAL-TIME EDUCATIONAL DIGITAL TWIN

- Playable 2.5D Campus World built with Phaser 3 & React
- Real-Time WebRTC Virtual Classrooms with Server-Enforced Audio Isolation
- Intelligent A* Pathfinding & Voice-Enabled AI Assistant
- **Presented By**: Akshat Lomas / Priyanshu Jain | **B.Tech** CSE (AI)
- **Institute**: Indian Institute of Information Technology Bhopal (IIIT Bhopal)

WHY DID WE BUILD THIS?

- **Transit Campus Navigation**: New students and visitors find it difficult to locate campus blocks, CCF labs, and departments.
- **Traditional Online Classes Lack Immersion**: Zoom and Google Meet do not provide the feeling of being in a real campus. They mostly show a simple grid of participants.
- **Audio Leakage Issue**: Virtual classrooms often lack proper control over background noise and audio separation.
- **Accessibility Awareness**: Students with disabilities (PwD) do not have easy access to real-time visual information about ramps and accessible campus facilities.

OUR SOLUTION: VIRTUAL DIGITAL TWIN

- **Dual-Layer Architecture**:
  - **Spatial Layer**: Phaser 3 powered 2.5D campus world (1600 × 1400 px, 60 FPS).
  - **Communication Layer**: Real-time WebRTC audio/video mesh + WebSockets.
- **Zero-Crash Local Deployment**: Runs out of the box in Demo Mode (bundled in-memory datasets) with optional MongoDB persistence.
- **Zero-Cost Operation**: AI Assistant aur WebRTC audio 100% are executed locally, without any expensive paid API .

TECHNOLOGY STACK & FRAMEWORKS

Frontend: React 18, Vite, Tailwind CSS, Lucide Icons

- **Game Engine**: Phaser 3 (Arcade Physics, Tilemaps, Cameras, Sprites)
- **Backend & APIs**: Node.js, Express.js (REST Architecture)
- **Real-Time Communication**: Native WebSockets (RFC 6455) & WebRTC MediaStreams
- **Token Specification**: LiveKit Access Token Schema (HS256 JWT)
- **Database**: Dual Mode — In-Memory JSON (Demo) & MongoDB (Production)

PLAYABLE CAMPUS & INTELLIGENT NAVIGATION

- **Avatar Controls**: 4-directional movement (WASD/Arrows), Shift Sprint (320 px/s), click-to-move.
- *A Search Algorithm**: Grid-based path calculation avoiding building colliders.
- **Navigation Metrics**: Real-time distance in meters & walking ETA in minutes.
- **Dynamic Re-routing**: If the player moves more than 120 pixels away from the planned path, the system automatically recalculates a new route.
- **Campus Environment**: Dynamic Day/Night lighting shader, glowing streetlights, central plaza fountain, and 4 interactive NPCs.

REAL-TIME CLASSROOMS WITH AUDIO ISOLATION

- **Server-Enforced Room Isolation**:
- **Outside Campus**: Outdoor ambient sound is audible.
- **Inside Classroom A (**`lh101`**)**: Outside campus sound auto-mutes . only Classroom A participants audio is audible.
- **Classroom B (**`lh102`**)**: zero audio is leaked to Classroom A (Zero Audio Bleed).
- **Clean Lifecycle Teardown**: When the user exits the classroom, camera, microphone, and screen-sharing tracks are automatically stopped using `track.stop()`, and the campus ambience is restored.
- **Role-Based Dashboards**:
- **Teacher**: Start/End session, Mute student, Screen share, Attendance CSV export.
- **Student**: Mic/Cam consent, Raise Hand (✋ chime), Q&A doubts queue, Chat.

COLLABORATIVE SMARTBOARD & VOICE AI

- **4K Synchronized Whiteboard**:
- HTML5 Canvas drawing (Pen, Eraser, Color Palette, Stroke Widths).
- Real-time WebSocket sync across all connected students.
- **Teacher Lock**: The teacher can lock drawing access so that students can only view the whiteboard.Users can export the whiteboard as a PNG image with a single click.
- **Voice-Enabled AI Assistant**:
- NLP intent classifier for campus directions, timings, and faculty queries.
- **Text-to-Speech (📣)**: Using Web Speech API integration, the AI guide provides spoken directions to help users navigate the campus.
- 100% Free local fallback mode.

DATASET, INCLUSIVITY & TESTING

- **Institutional Dataset**: 14 core verified landmarks (CCF Lab, ECE Block, Central Library, Food Court, Hostels).
- **100% PwD Friendly Accessibility**:
- Step-free 1:12 slope wheelchair entrance ramps.
- Classrooms me Row 1 dedicated wheelchair workstations.
- Audio induction loops and tactile floor markers.
- **Automated Verification**:
- 16/16 Unit & Integration Tests passed.
- Tests verify LiveKit JWT generation, security roles, room isolation, and A* path optimality.

CONCLUSION & LIVE DEMONSTRATION

- **Project Achievements**:
- Browser-based Metaverse running at 60 FPS without high-end GPU.
- Zero external cloud cost; fully runnable on a standard student laptop.
- High utility for actual institutional orientation and hybrid classes.
- **Future Roadmap**: Multi-avatar presence on the open campus grid, integration with official institute noticeboards.